

Regal Jumping Spider (*Phidippus regius*)

Difficulty: Easy to Intermediate

Regal jumping spiders are among the largest jumping spiders in eastern North America. Adult males average about 12 mm and females about 15 mm, although large females can approach 22 mm. They are visual, daytime hunters rather than web-trapping spiders, stalking live prey and using a silk safety line when they jump.

Phidippus regius occurs in the southeastern United States, the Bahamas, and the Greater Antilles and is especially common in peninsular Florida. Wild regals use open fields, light woodland, palms, shrubs, tree trunks, and even building walls. They build dense silk retreats for sleeping, molting, and egg-laying, usually in protected elevated locations.

Jumping-spider care is unusually prone to confident but conflicting advice. For this guide, Red Rocks prioritizes the points supported by natural history, published laboratory work, and broad agreement among experienced *Phidippus* keepers: strong cross-ventilation, reliable access to tiny drinking droplets, bright daytime light without direct sun, appropriately sized live prey, an undisturbed upper retreat, and individual housing. These recommendations are for *P. regius* and should not be generalized to every salticid species.

Shopping List

- Vertical front-, side-, or bottom-opening enclosure; about 4" x 4" x 8" is a practical adult minimum, with 5" x 5" x 8" to 6" x 6" x 9" excellent for large active females
- Smaller escape-proof grow-out cup or enclosure for spiderlings and small juveniles
- Fine cross-ventilation on at least two sides; openings must be smaller than the spider and its feeder insects
- Bright white LED or other low-heat visible light on a timer for a 12-hour day/night cycle
- Digital thermometer; a small hygrometer can be useful but should not replace direct observation of drying and ventilation
- Fine-mist spray bottle or pipette for placing tiny drinking droplets on one wall
- Cork, thin branches, textured climbing surfaces, foliage, and an elevated hide or leaf for hammock anchors
- Thin paper towel, bare bottom, or a shallow layer of clean coco/soil substrate depending on the setup
- Flightless fruit flies for small juveniles; bottle/house flies, small roaches, small crickets, and similar cultured feeders for larger spiders
- Feeding cup or smooth-sided feeder dish for worms/roach nymphs when useful
- Soft paintbrush or catch cup for safe transfers
- Optional magnifier for checking molt status, mature sex characteristics, and condition without handling

Housing

Regal jumping spiders are primarily arboreal and should be housed in a vertical enclosure with usable surfaces from bottom to top. An average adult can be maintained well in a properly furnished enclosure around 4" x 4" x 8"; larger adults, especially females, benefit from roughly 5" x 5" x 8" to 6" x 6" x 9" or similar. More space is not automatically better for a tiny spiderling: young spiders should be moved up gradually so they can reliably locate prey and water.

Cross-ventilation is more important than trying to hold a high humidity number. Use vents on opposing sides or multiple sides, not only a few holes in the lid. Every opening, cable gap, hinge, or door seam must be small enough to contain both the spider and the feeder insects.

Use a front-, side-, or bottom-opening enclosure whenever possible. Regals commonly build sleeping and molting hammocks at the top of the enclosure. A top-opening lid that repeatedly tears the retreat

can force the spider to rebuild and may be especially disruptive during premolt, molting, egg guarding, or old age.

Lighting

Jumping spiders are visual predators and should receive a clear period of bright, diffuse visible light during the day, followed by real darkness at night. A roughly 12-hour light cycle works well. Published salticid research demonstrates sophisticated vision even at low light, but active hunting and normal visual behavior occur under daytime conditions; a low-heat white LED is an easy way to provide a predictable day period indoors.

UV perception is widespread in jumping spiders, but that does not establish a captive UVB requirement for *P. regius*. Controlled husbandry data showing that UVB improves health or molting in regals are lacking, so UVB is not part of our standard recommendation. Never place a small enclosure in direct sunlight or under a high-output reptile lamp: tiny clear enclosures can overheat dangerously fast.

Temperature

Normal indoor temperatures suit this species well. A practical daytime target is about 72-80°F (22-27°C), with a modest nighttime drop into the upper 60s or low 70s acceptable for healthy animals. *P. regius* has been maintained successfully in research at ordinary room temperature, and captive care does not require a hot basking zone.

Avoid prolonged temperatures above roughly 82-85°F inside a small enclosure, especially if airflow is limited. If the room is genuinely too cool, warming the room is safer than aiming a heat lamp at the spider. Any supplemental heat should be indirect, thermostatically controlled, and arranged so the spider can move away from it.

Humidity & Water

The most important hydration goal is access to drinkable droplets without creating stagnant, constantly wet conditions. For most captive *P. regius*, roughly 50-70% relative humidity with excellent airflow is a useful working range, but tiny enclosure hygrometers can be inaccurate. Watch how quickly droplets and surfaces dry instead of chasing a single number.

As a starting routine, lightly mist one wall or a few leaves every 1-3 days so fine droplets are available for drinking, then let most surfaces dry. Very dry heated homes may need smaller droplets more often; humid rooms may need less. Do not soak the substrate, fog the entire enclosure, or spray the spider directly. During premolt, pay extra attention to hydration but do not blast water into the sealed hammock.

Open water bowls are unnecessary and are a greater drowning hazard for spiderlings and small juveniles than fine droplets. If an adult needs emergency hydration, offer a small droplet on a cotton swab or enclosure wall rather than forcing the mouthparts into water. Never use honey, sugar water, or fruit as routine hydration.

Substrate

Regal jumping spiders do not need deep substrate. A bare floor, paper towel, or a shallow layer of clean coco fiber/soil can all work. Simple floors make it easier to find waste, feeder insects, and fallen molts; a thin naturalistic substrate can buffer moisture but should never remain soggy.

Avoid sharp gravel, loose fibers that can entangle legs, pesticide-treated plant material, sticky tapes, exposed adhesive, and uncured paints/resins. Bioactive setups are possible, but for a single small spider they add complexity and can make prey and health monitoring harder.

Food

P. regius is a generalist arthropod predator. Natural-history work records smaller immatures feeding heavily on flies and older spiders taking larger insects including orthopterans, hemipterans, and

moths. In captivity, use clean cultured prey and rotate feeder types rather than relying on one insect for the spider's entire life.

Spiderlings and small juveniles: *Drosophila melanogaster* or *D. hydei*, offered every 1-2 days or according to abdomen size. Larger juveniles: larger flies, small roaches, or carefully sized crickets every 2-3 days. Adults: bottle flies, houseflies, small roaches, small crickets, or similar prey about every 4-7 days as a starting point. Feeding should ultimately follow body condition and activity, not a rigid calendar.

Choose prey the spider can overpower safely; when uncertain, choose smaller prey. Flies are excellent staples because they encourage natural visual hunting and are relatively low risk. Crickets and other biting feeders should be removed promptly if ignored, and no live prey should remain with a spider that is sealed into a molt hammock. Mealworms can be offered in moderation in a smooth feeding cup; waxworms are an occasional high-calorie treat, not a staple.

Gut-load feeder insects when practical. Routine calcium or reptile-vitamin dusting is not established as necessary for jumping spiders and is not recommended as a default. Never feed wild-caught insects because pesticide exposure, parasites, and defensive chemicals are unpredictable. Do not use honey, fruit, or sugar-water "boosts" as a substitute for appropriate prey and clean water.

Molting

Molting is the period when otherwise simple jumping-spider care becomes most delicate. Before a molt, a spider may stop eating, become less active, and build a noticeably thicker hammock. These changes can last days or longer and are not a reason to tear open the retreat to check on the spider.

Once a spider seals itself in, remove any dangerous live prey and leave the enclosure stable. Continue providing small water droplets away from the hammock as needed, but do not mist directly into the retreat. Do not pick up, rotate, or repeatedly move the enclosure unless there is an emergency.

After the spider molts, wait until it emerges willingly and its body and chelicerae have hardened before offering risky prey or handling. Small juveniles may be ready sooner, while a large adult can need several days. If the spider remains hidden, let the spider set the schedule rather than forcing it out.

Decor & Enrichment

Decor should create routes and choices, not just fill space. Use cork, twigs, thin branches, textured vertical surfaces, foliage, and elevated ledges so the spider can stalk, jump, hide, and anchor silk. Leave open sight lines between some perches because visual hunting is a major part of normal behavior.

Provide more than one secure upper retreat location. As spiders age, their grip can weaken; add intermediate perches and foliage that reduce long fall distances. Every decoration should be firmly secured and free of sharp edges, exposed wire, sticky adhesive, or loose fibers.

Cleaning & Maintenance

Remove prey remains, frass, dead feeders, and obvious mold as needed. Full enclosure sterilization is rarely necessary for a healthy established spider and can destroy important silk anchor points. Clean around the hammock whenever possible rather than removing it simply because it looks messy.

Never use insecticides, aerosol cleaners, fragrances, essential oils, flea/tick products, or household sprays near the enclosure. Spiders are highly sensitive to many chemicals designed to kill arthropods. Keep new spiders and feeder cultures away from areas recently treated for pests.

Cohabitation

House one regal jumping spider per enclosure. Jumping spiders are predators, not animals that need companionship, and cannibalism is a real risk. Do not keep pairs together as pets and do not add

other spiders or “cleanup” arthropods large enough to interact with the jumper. Breeding introductions are brief, supervised events for experienced keepers only.

Egg Sacs & Reproduction

Mature females may build a dense egg nest even when no male is present. An unmated female can lay infertile eggs, while a female that has mated previously can store sperm and produce more than one fertile egg sac from a single mating. If a female’s history is unknown, treat a new egg sac as potentially fertile until proven otherwise.

Do not tear open a guarded egg nest or repeatedly disturb the female. A fertile *P. regius* sac can produce dozens to hundreds of spiderlings, and a female may produce multiple sacs. Raising that many slings requires tiny escape-proof containers, continuous fruit-fly cultures, and a plan for separation and placement; contact an experienced breeder before the spiderlings disperse if you are not prepared for that commitment.

Common Issues

- Dehydration: a markedly shrunken abdomen, lethargy, or poor coordination can accompany dehydration. Offer safe droplets and correct the enclosure gradually rather than flooding it.
- Overfeeding: an extremely distended abdomen increases the consequences of a fall and is not a sign that the spider needs another meal. Let the abdomen slim before feeding again.
- Failed or difficult molt: inadequate hydration, disturbance, injury, age, or underlying weakness may contribute. Do not pull an attached molt from the spider; seek experienced arachnid help if the spider is trapped.
- Chemical exposure: tremors, uncontrolled movement, repeated falling, or sudden neurologic-looking behavior can follow pesticide or household chemical exposure. Move the enclosure away from possible contamination and avoid adding unproven home remedies.
- Poor grip in old age: mature spiders can lose climbing ability. Add lower routes and soft foliage, reduce fall height, and offer easier prey rather than repeatedly rehousing the spider.
- Persistent refusal to eat: premolt, maturity, egg guarding, low temperature, dehydration, oversized prey, or senescence can all reduce appetite. Evaluate context before assuming illness.

Hot Tip: Handling

Handling is optional enrichment for the keeper, not a requirement for the spider. Let the spider choose to step onto a clean hand or catch cup; never pinch, grab, blow on, or chase it. Handle low over a table, bed, or other safe surface because a fall can rupture a well-fed abdomen.

Wash and thoroughly rinse hands before contact, then make sure they are free of soap residue, sanitizer, lotion, nicotine, cleaners, and pet pesticides. A defensive bite is uncommon but possible, especially if the spider is restrained. If the spider repeatedly retreats or displays defensively, end the interaction.

Care Standards Note

This guide is intentionally species-specific and conservative. *P. regius* comes from a broad geographic range, and individual spiders vary with age, sex, locality, room climate, and life stage. Where controlled captive-husbandry trials do not exist, recommendations are based on published *P. regius* natural history and laboratory methods together with points that are broadly consistent across experienced captive breeders. We intentionally do not recommend constant high humidity, routine UVB, standing water for small spiders, sugar-water “supplements,” forced handling, or cohabitation as default care.